Sahaj Bhadja

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Summary

Software engineer with 2+ years building **high-performance C++/Python systems**, **simulation-oriented tooling**, and **distributed validation workflows** for autonomy-style, safety-critical product environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **Python/C++ data services** and tooling for 500+ users, reducing end-to-end latency from **1.2s to 350ms** through performance profiling and systems-level optimization.
- Implemented **fault-tolerant orchestration and validation** for 5K+ monthly jobs, improving reliability and accelerating root-cause analysis across cross-functional software pipelines.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Developed **testable product infrastructure** handling 10K+ daily requests with React/Flask, translating requirements into robust APIs/components for rapid iteration and deployment.
- Improved **CI/CD automation and regression quality** with scripted checks and defect triage, reducing release turnaround by **60%** while maintaining stability in production.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Built **simulation-scale experimentation pipelines** over 100K+ trajectories, enabling reproducible benchmark execution and quantitative evaluation of autonomy-related behavior patterns.
- Created **debugging diagnostics and technical summaries** that translated research outcomes into actionable engineering improvements across multi-team development efforts.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **distributed backend services** with Python/Java, AWS Glue, and S3, improving response times by **30%** and supporting reliable data operations at scale.
- Maintained **20+ automated test suites** on Linux/Windows with Jenkins, PyTest, and Mockito, improving pipeline efficiency by **68%** and reducing defect recurrence.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Implemented **MERN + WebSocket modules** for eight workflows and 2M+ synchronized records, improving consistency, responsiveness, and reliability of real-time platform interactions.
- Automated **containerized deployment pipelines** with Docker/Jenkins, reducing rollout time by **50%** through repeatable build, validation, and release processes.

Skills

Languages: **C++**, **Python**, Java, JavaScript/TypeScript, SQL, Bash
Systems: concurrency, multithreading, performance optimization, Linux/UNIX environments
Simulation & Autonomy: virtual scenario testing, synthetic data workflows, real-time behavior validation
Infrastructure: distributed services, CI/CD automation, Docker, Kubernetes basics, Git/GitHub
Engineering: architecture/design reviews, requirements translation, troubleshooting, cross-functional communication

Projects

CPU Scheduling Simulator Live | *C++, React.js, JavaScript*

- Implemented **FCFS, SJF, SRT, RR, and MLFQ algorithms** with runtime metrics and visualization, reducing debugging time by **20%** through deterministic scenario replay.

Autonomy Scenario Replay Tool | *C++, Python, JSON, OpenGL*

- Built a **high-performance scenario replay pipeline** for sensor/event streams with configurable timing controls, improving reproducibility and integration testing across simulated missions.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized REST processing service** with automated CI/CD validation, improving production reliability and operational continuity for large-scale workloads.